

Profile Summary Portfolio

AI/ML Engineer with 5 years of software engineering experience, specializing in production GenAI systems, LLM optimization, RAG, NLP-to-SQL, agentic AI, and model fine-tuning. Built and optimized AI systems across the full stack — from specialized models and retrieval pipelines to APIs, orchestration, evaluation, and GPU infrastructure. Proven experience reducing LLM cost and latency, improving concurrent serving, fine-tuning open-source models, and replacing general-purpose LLM components with specialized models for constrained tasks.

Work Experience

Advanced Micro Devices (AMD) — AI Software System Engineer (Jan 2026 - Present)

- Designed and developed an Agentic AI platform for GPU infrastructure support using Agno and MCP, integrating documentation retrieval, historical ticket search, tool execution, automated pull-request generation, and engineering escalation. Used by 10+ internal teams.
- Fine-tuned and evaluated LLMs for Digital Coworker applications on AMD GPU clusters, improving domain-specific response quality and RAG workflows.
- Developed automation and infrastructure tooling using Slurm, Salt, Ansible, and Spur supporting distributed LLM training and inference across infrastructure containing thousands of AMD Instinct MI355 GPUs.

PowerSchool — Software Engineer I (Apr 2023 – Jan 2026)

Promoted from Associate SE I & II

- **NLP-to-SQL:** Designed and implemented major components of PowerBuddy's production NLP-to-SQL analytics engine, covering semantic table/column selection, SQL generation, data retrieval, charts, and AI-generated insights.
- **LLM Optimization:** Built a specialized **LLaMA-based alternative to GPT** for constrained semantic-selection and query-generation tasks, reducing complete workflow latency from ~60 seconds to **~18 seconds typical (15–28s observed)** while substantially reducing model-serving cost.
- **Pipeline Optimization:** Independently optimized the existing GPT-based pipeline, reducing end-to-end latency from ~60 seconds to **10–23 seconds** and increasing tested concurrency from approximately **4 to 10 simultaneous users** on the same compute environment. Performance was independently validated internally.
- **LLM Fine-Tuning:** Fine-tuned LLaMA models for structured JSON/YAML generation using **SFT, PPO, DPO, and custom reward functions**, targeting structural validity and semantic consistency across SQL, chart, and insight-generation tasks.
- **Embedding Compression:** Developed a **Siamese MiniLM-based network** compressing 384-dimensional embeddings to 64D, achieving approximately **4× faster vector search and >60% lower embedding memory usage** while preserving semantic relevance.
- **Vector-Free RAG:** Built a lightweight semantic retrieval system using **NumPy and pandas** instead of a vector database, achieving **<150 ms retrieval across 10K+ documents** using precomputed embeddings and cosine similarity.

- **Streaming Structured Generation:** Built a fault-tolerant parser that reconstructs usable JSON from incomplete streaming LLM output, enabling progressive UI rendering.

PowerSchool — Associate Software Engineer I & II (Jul 2021 – Mar 2023)

- Developed enterprise applications using ASP.NET/C#, Angular, and SQL, contributing to ERP systems, student modeling, adaptive testing, and personalized content before transitioning internally to the AI/Data Science team.

Personal AI Systems & Projects | 30+ Public Repositories & Experiments

- **Cortana++ (In Progress):** C++/CUDA-based ML engine built from scratch to support tensors, broadcasted matmul, and Dense layers, targeting low-memory environments.  [GitHub](#)
- **GreedyContext:** Built a semantic graph using sentence embeddings + greedy traversal to shrink LLM context by 50–90%, improving latency and reducing cost.  [GitHub](#)
- **DeepFake Inference (OpenCL):** Built a DeepFake pipeline using OpenCL on AMD GPUs, manually implementing ANN inference without external ML libraries.  [GitHub](#)
- **Unity3D Self-Driving Simulation:** Simulated PPO agents navigating GPS-defined tracks using a custom-designed car and road system in Blender + Unity.  [GitHub](#)
- **Real-Time Face Recognition System:** Used OpenCV + TensorFlow to mark attendance via facial authentication and store logs in MySQL.  [GitHub](#)

Core Competencies

- LLM Fine-Tuning (SFT, PPO, DPO)
- Retrieval-Augmented Generation (RAG)
- Low-Level ML Infra (C++/CUDA/OpenCL)

Technical Skills

- **Languages:** Python, C++, Java, C#, JavaScript
- **ML Frameworks:** PyTorch, TensorFlow, TRL, OpenCV
- **GPU Programming:** CUDA Kernels, OpenCL, Numba
- **Data & Backend:** Snowflake, SQL, MongoDB, Pandas
- **Simulation & 3D:** Unity3D, Blender
- **Web Technologies:** React.js, Angular
- **MLOps:** AWS SageMaker, EC2, Docker, K8s, MD GPU Clusters, Slurm, Salt, Ansible,